Logistic Regression and Naïve Bayes

Lecture 03

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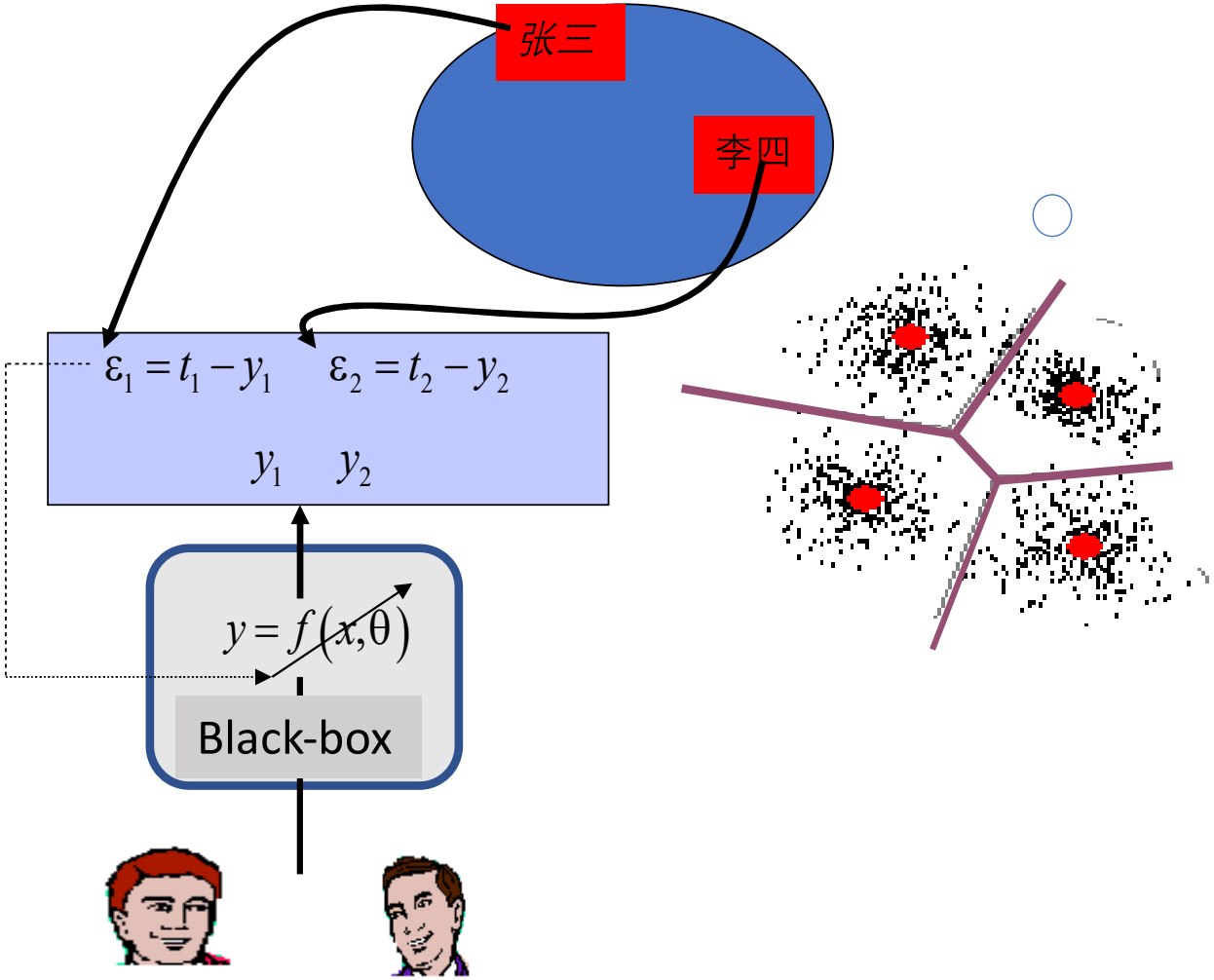
Outline

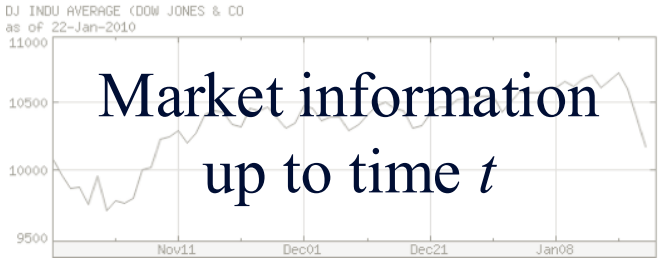
- **Logistic regression**
- Naïve Bayes
- Their Relationship

Supervised Learning



Supervised Learning

- Regression



Market information up to time t

Share Price
"$ 24.50"

Continuous Labels
Regression

- Classification

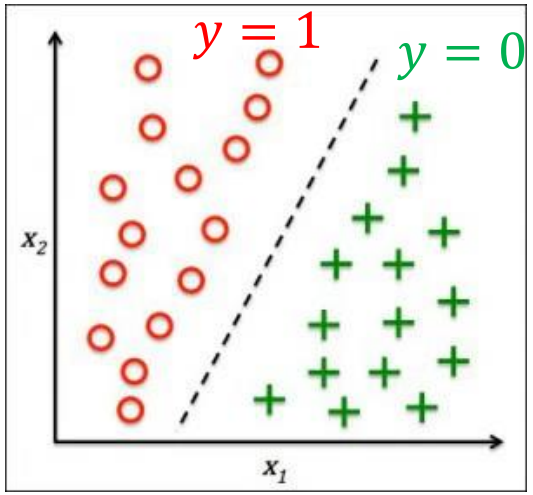
Feature Space $\mathcal{X}$

Words in a document

Label Space $\mathcal{Y}$

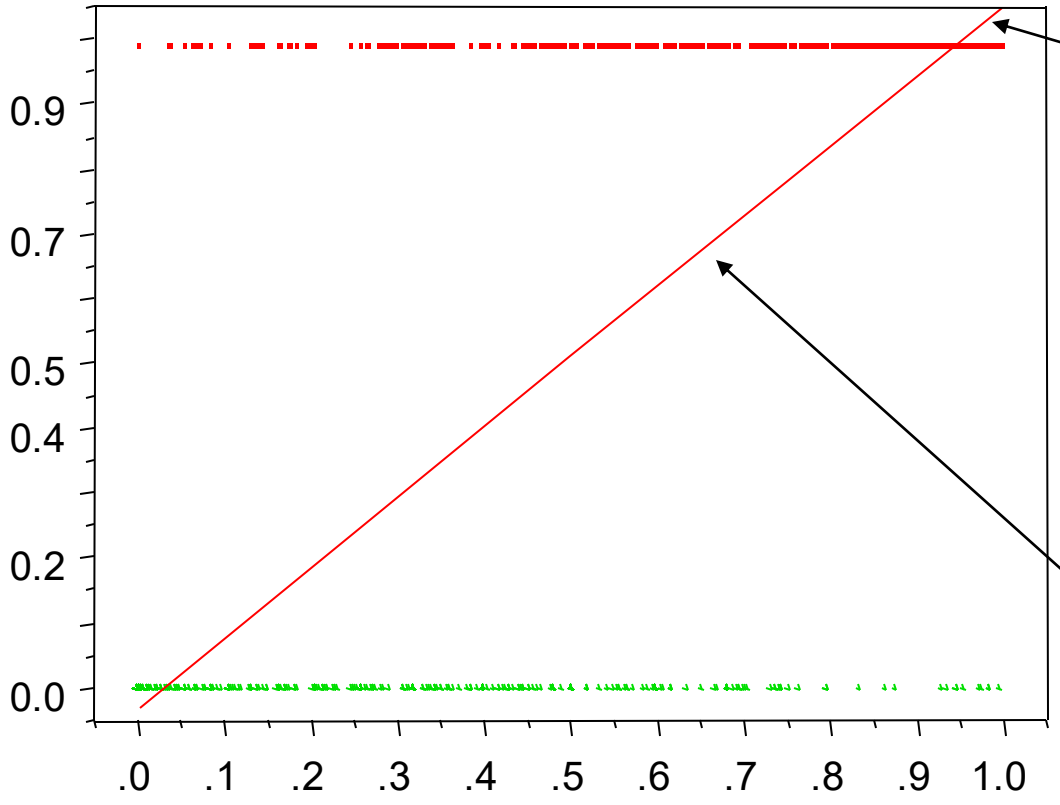
"Sports"
"News"
"Science"
…

Discrete Labels
Classification



Why not Linear Regression?

- For classification, Y only takes on values of 0 and 1.



How do we interpret values greater than 1 and smaller than 1?

How do we interpret values of Y between 0 and 1?

Problems

- The regression line $a_0 + a_1 x$ can take on any value between negative infinity and positive infinity.

- In the classification problem, output Y can only take on two possible values: 0 or 1.

- Therefore the regression line almost always predicts the wrong value for output Y in classification problems

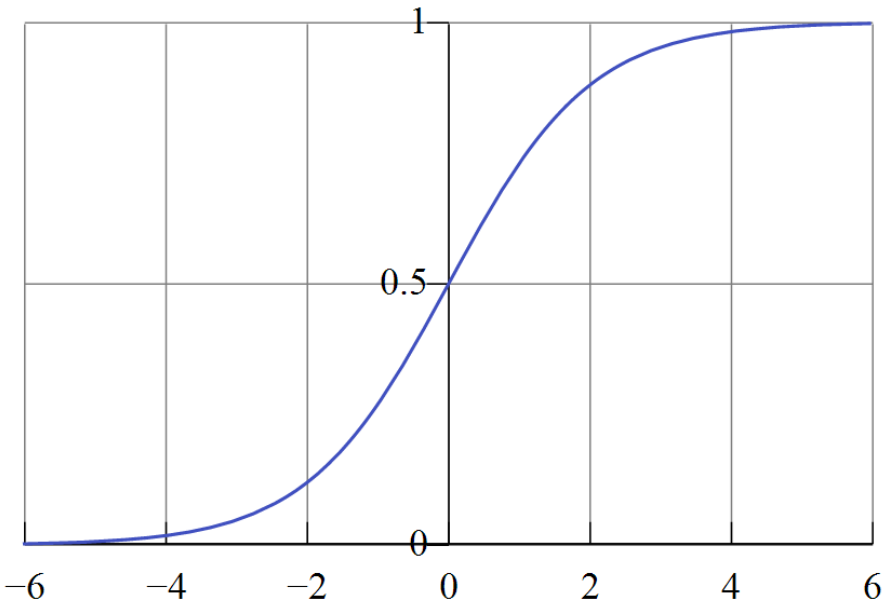
Solution: Use the logistic function

- Instead of trying to predict Y, let's try to predict $P(Y = 1)$, which is continuous number between 0 and 1.

- Model $P(Y = 1)$ using a function that gives outputs between 0 and 1 → logistic function!

$$\text{Sigmoid}(x) = \text{Logistic}(x) = \frac{1}{1 + e^{-x}}$$

$$= \frac{e^x}{e^x + 1} = \frac{1}{2} + \frac{1}{2}\tanh\left(\frac{x}{2}\right)$$



Logistic regression

- The model

$$P(Y=1|X=x)=P(Y=1|x)=\frac{e^{a_0+a_1x}}{1+e^{a_0+a_1x}}=\frac{1}{1+e^{-(a_0+a_0x)}}$$

- The Euler's number $e=2.71828\ldots$
- No matter what values a_0, a_1 or x take, $P(Y=1|x)$ will have values between 0 and 1.

$$\log\left(\frac{P(Y=1|x)}{P(Y=0|x)}\right)=\log\left(\frac{P(Y=1|x)}{1-P(Y=1|x)}\right)=a_0+a_1x$$

Log-odds or logit transformation of $P(Y=1|x)$ gives a linear model.

Interpreting the coefficient a_1

$$P(Y = 1|x) = \frac{1}{1 + e^{-(a_0 + a_0 x)}}$$

- Not very easy with logistic regression → We are predicting $P(Y = 1|x)$, not Y directly
  - If $a_1 = 0$ → no relationship between Y and X.
  - If $a_1 > 0$ → when X gets larger so does the probability that $Y = 1$.
  - If $a_1 < 0$ → when X gets larger, the probability that $Y = 1$ gets smaller
  - How much bigger or smaller depends on where we are on the slope

Logistic regression with multiple variables

- In real applications, the input X usually has multiple features, i.e., $\boldsymbol{x} = [x_1, x_2, \dots, x_n]$
- The logistic regression naturally extends as:

$$P(Y = 1|\boldsymbol{x}) = \frac{e^{a_0 + a_1 x_1 + \cdots a_n x_n}}{1 + e^{a_0 + a_1 x + \cdots a_n x_n}}$$
$$= \frac{e^{a_0 + \boldsymbol{a}^T \boldsymbol{x}}}{1 + e^{a_0 + \boldsymbol{a}^T \boldsymbol{x}}}$$
$$= \frac{1}{1 + e^{-(a_0 + \boldsymbol{a}^T \boldsymbol{x})}}$$

Without loss of generality, we may have $P(Y = 1|\boldsymbol{x}) = \dfrac{1}{1 + e^{-\boldsymbol{a}^T \boldsymbol{x}}}$

where $\boldsymbol{x} = [1, x_1, x_2, \dots, x_n]^T$, $\boldsymbol{a} = [a_0, a_1, a_2, \dots, a_n]^T$.

Maximum Likelihood (ML) learning

- **Questions**
  - What is objective function for logistic regression?
  - Can it be mean square error (MSE), which has been used in linear regression?

Training Data set

$$\{(\boldsymbol{x}_t, y_t)\}_{t=1}^{N}$$

$$X_N = \{\boldsymbol{x}_t\}_{t=1}^{N}$$

$$Y_N = \{y_t\}_{t=1}^{N}$$

- **Likelihood function**

$$P(Y_N|X_N,\theta) = \prod_{t=1}^{N} P(y_t|x_t,\theta) = \prod_{t=1}^{N} \sigma(\boldsymbol{x}_t)^{y_t}\big(1-\sigma(\boldsymbol{x}_t)\big)^{1-y_t}$$

Logistic function

$$\sigma(\boldsymbol{x}_t) = \frac{1}{1+e^{-\boldsymbol{a}^T\boldsymbol{x}}}$$

- **Log-Likelihood function**

$$\mathcal{L}(Y_N|X_N,\theta) = \log P(Y_N|X_N,\theta) = \sum_{t=1}^{N}\{y_t\sigma(\boldsymbol{x}_t) + (1-y_t)(1-\sigma(\boldsymbol{x}_t)\}$$

Cross-entropy

parameters

$$\theta = \{\boldsymbol{a}\}$$
$$= \{a_0, a_1, \dots, a_n\}$$

Logistic regression with more than two classes

- Logistic regression is easily generalized to classify multiple classes:

$$P(Y = k|X = \boldsymbol{x}) = \frac{e^{a_{0k}+a_{1k}x_1+\cdots+a_{nk}x_n}}{1+\sum_{i=1}^{C} e^{a_{0i}+a_{1i}x_1+\cdots+\beta_{pi}x_n}}$$

C is the number of classes

- In the matrix form:

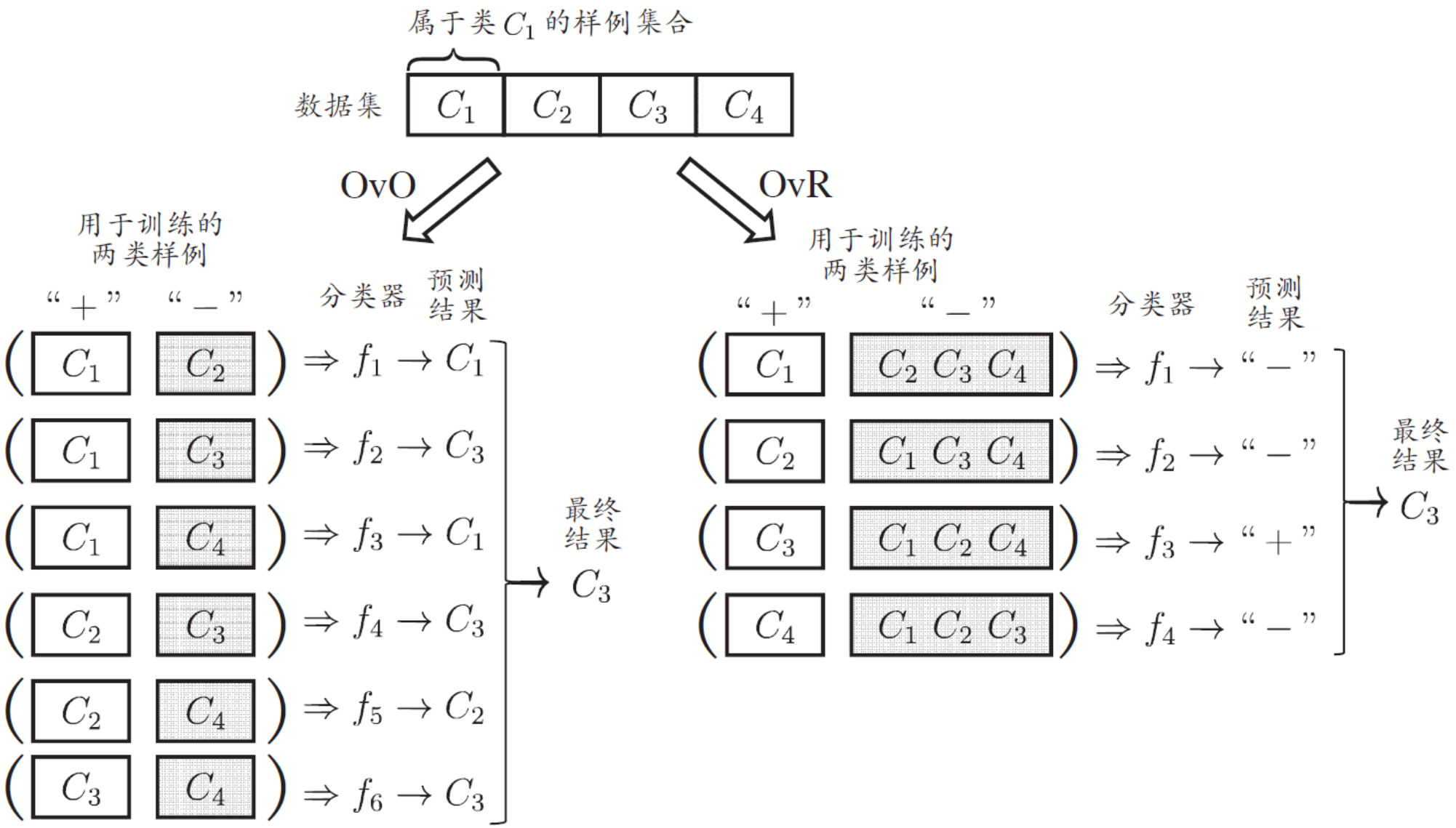
$$P(Y = k|X = \boldsymbol{x}) = \frac{e^{\boldsymbol{a}_k^T\boldsymbol{x}}}{1+\sum_{i=1}^{C} e^{\boldsymbol{a}_k^T\boldsymbol{x}}}$$

where $\boldsymbol{x} = [1, x_1, x_2, \dots, x_n]^T$, $\boldsymbol{a}_k = [a_{0k}, a_{1k}, a_{2k}, \dots, a_{nk}]^T$

From one class to multiple classes

- Multi-classification methods
  - Extend the binary classification to tackle the multi-classification problem
  - Use the split-strategy
- The split strategy
  - Split the multi-classification into a set of binary classification problems
  - Train a classifier for every binary classification problem
  - Ensemble the results from every binary classifier to obtain the multi-classification results
  - How to split?
    - One vs. One (OvO)
    - One vs. Rest (OvR)
    - Many vs. Many (MvM)

Comparing the two splitting ways



Comparing the two splitting ways

- **One vs. One**
  - The time and space for training N(N-1)/2 classifiers is expensive.
  - Training a classifier only requires data from two classes, and thus the training time is short.
- **One vs. Rest**
  - The time and space for training N classifiers is cheap.
  - Training a classifier requires data from all classes, and thus the training time is long.
- The prediction performance depends on the specific data distribution.
- The two strategies are the same for most cases.

Outline

- Logistic regression
- **Naïve Bayes**
- Their Relationship

Naïve Bayes（朴素贝叶斯）

- **核心思想**
  - 利用已知的训练数据（特征和对应的类别标签），基于贝叶斯定理，计算一个新样本属于各个类别的概率，并将它分到概率最大的那个类别中。
- **特征条件独立性假设**（Feature Independence Assumption）
  - 即假设给定样本的类别时，所有特征（属性）之间是相互独立的。
  - 这个假设简化了计算，但在现实中特征之间往往存在相关性，所以称之为“朴素”
- **学习目标**
  - 给定一个新的样本特征向量 $X = [x_1, \dots, x_n]$，计算它属于每个类别 c 的概率 $P(Y = c|X)$，并选择概率最大的那个类别作为预测结果：

$$\hat{c} = \arg\max_{c} P(Y = c|X)$$

$$P(Y = c|X) = \frac{P(X|Y = c)P(Y = c)}{P(X)}$$

$$P(X|Y = c) = \prod_{i=1}^{n} P(x_i|Y = c)$$

Naïve Bayes (NB) for binary classification

$$\hat{c} = \arg\max_{c} P(Y = c|X)$$

$$X = [x_1, \dots, x_n]$$

- Binary classification: $Y \in \{0,1\}$
- Decision is made by comparing the log odds:

$$\log \frac{P(Y = 1|X)}{P(Y = 0|X)} = \log \frac{P(X|Y = 1)P(Y = 1)}{P(X|Y = 0)P(Y = 0)}$$

- Based on the feature independence assumption, i.e., $P(X|Y = c) = \prod_{i=1}^{n} P(x_i|Y = c)$, we have

$$\log \frac{P(Y = 1|X)}{P(Y = 0|X)} = \sum_{i=1}^{n} \log \frac{P(x_i|Y = 1)}{P(x_i|Y = 0)} + \log \frac{P(Y = 1)}{P(Y = 0)}$$

- Under certain assumptions, NB is equivalent to logistic regression:

$$\log \frac{P(Y = 1|x)}{P(Y = 0|x)} = a_0 + a_1 x$$

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Naïve Bayes (NB) vs Logistic Regression (LR)

- When the features $x_i|Y$ is distributed according to an exponential family distribution, and satisfies the feature independence assumption, NB is equivalent to LR on the decision boundary.

- The exponential family distributions:

$$p(\mathbf{x}|\boldsymbol{\eta}) = h(\mathbf{x})g(\boldsymbol{\eta})\exp\left\{\boldsymbol{\eta}^{\mathrm{T}}\mathbf{u}(\mathbf{x})\right\}$$

Bernoulli distribution

$$\begin{aligned} u(x) &= x \\ h(x) &= 1 \\ g(\eta) &= \sigma(-\eta) \end{aligned}$$

multinomial distribution

$$\begin{aligned} \mathbf{u}(\mathbf{x}) &= \mathbf{x} \\ h(\mathbf{x}) &= 1 \\ g(\boldsymbol{\eta}) &= \left(1 + \sum_{k=1}^{M-1} \exp(\eta_k)\right)^{-1} \end{aligned}$$

Gaussian distribution

$$\begin{aligned} \boldsymbol{\eta} &= \begin{pmatrix} \mu/\sigma^2 \\ -1/2\sigma^2 \end{pmatrix} \\ \mathbf{u}(x) &= \begin{pmatrix} x \\ x^2 \end{pmatrix} \\ h(\mathbf{x}) &= (2\pi)^{-1/2} \\ g(\boldsymbol{\eta}) &= (-2\eta_2)^{1/2}\exp\left(\frac{\eta_1^2}{4\eta_2}\right) \end{aligned}$$

Exponential family

https://en.wikipedia.org/wiki/Exponential_family

A single-parameter exponential family is a set of probability distributions whose probability density function (or probability mass function, for the case of a discrete distribution) can be expressed in the form

$$f_X(x \mid \theta) = h(x)\ \exp[\eta(\theta) \cdot T(x) - A(\theta)]$$

where $T(x)$, $h(x)$, $\eta(\theta)$, and $A(\theta)$ are known functions. The function $h(x)$ must be non-negative.

An alternative, equivalent form often given is

$$f_X(x \mid \theta) = h(x)\, g(\theta)\ \exp[\eta(\theta) \cdot T(x)]$$

or equivalently

$$f_X(x \mid \theta) = \exp[\eta(\theta) \cdot T(x) - A(\theta) + B(x)].$$

In terms of log probability,

$$\log(f_X(x \mid \theta)) = \eta(\theta) \cdot T(x) - A(\theta) + B(x).$$

Note that $g(\theta) = e^{-A(\theta)}$ and $h(x) = e^{B(x)}$.

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